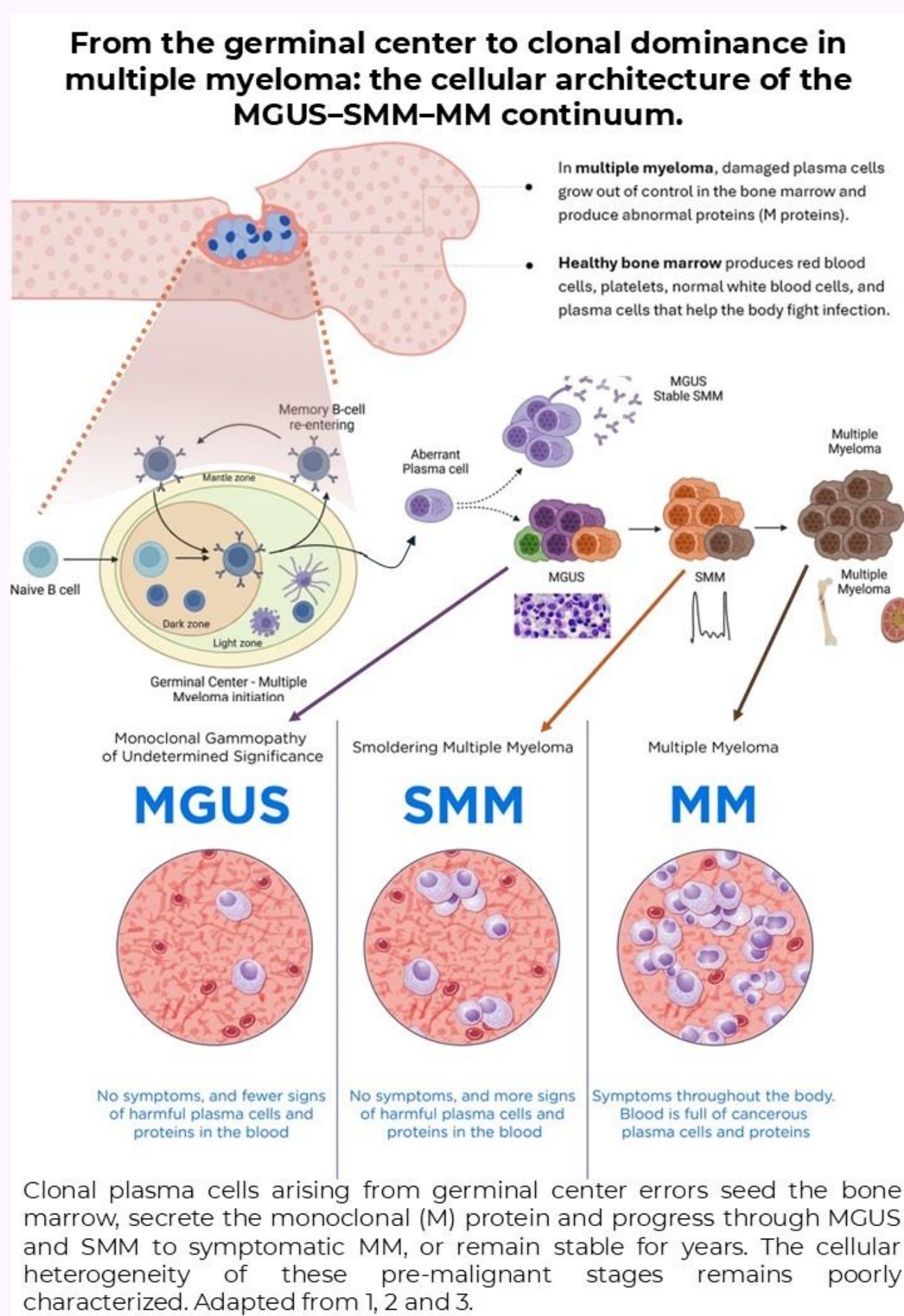


When Does Myeloma Really Begin? *Single-Cell Evidence That Transformation Precedes Diagnosis.*

Tiago Fernando Chaves^{1,2,*}, Juliane Heyde dos Santos^{1,2}, Taline Maria Fontana^{1,2}, Ricardo Henrique Döring^{1,2}, Bruna Terezinha Magnabosco Ferreira da Cruz², Juliana Dal-Ri Lindenau¹, Daniel Pacheco Bruschi², Yara Costa Netto Muniz¹.

¹Laboratório de Polimorfismos Genéticos (LAPOGE). Departamento de Biologia Celular, Embriologia e Genética, Universidade Federal de Santa Catarina (UFSC);
²Laboratório de Citogenômica Evolutiva. Departamento de Genética, Universidade Federal do Paraná (UFPR). *E-mail: chaves.t.f@posgrad.ufsc.br

Introduction

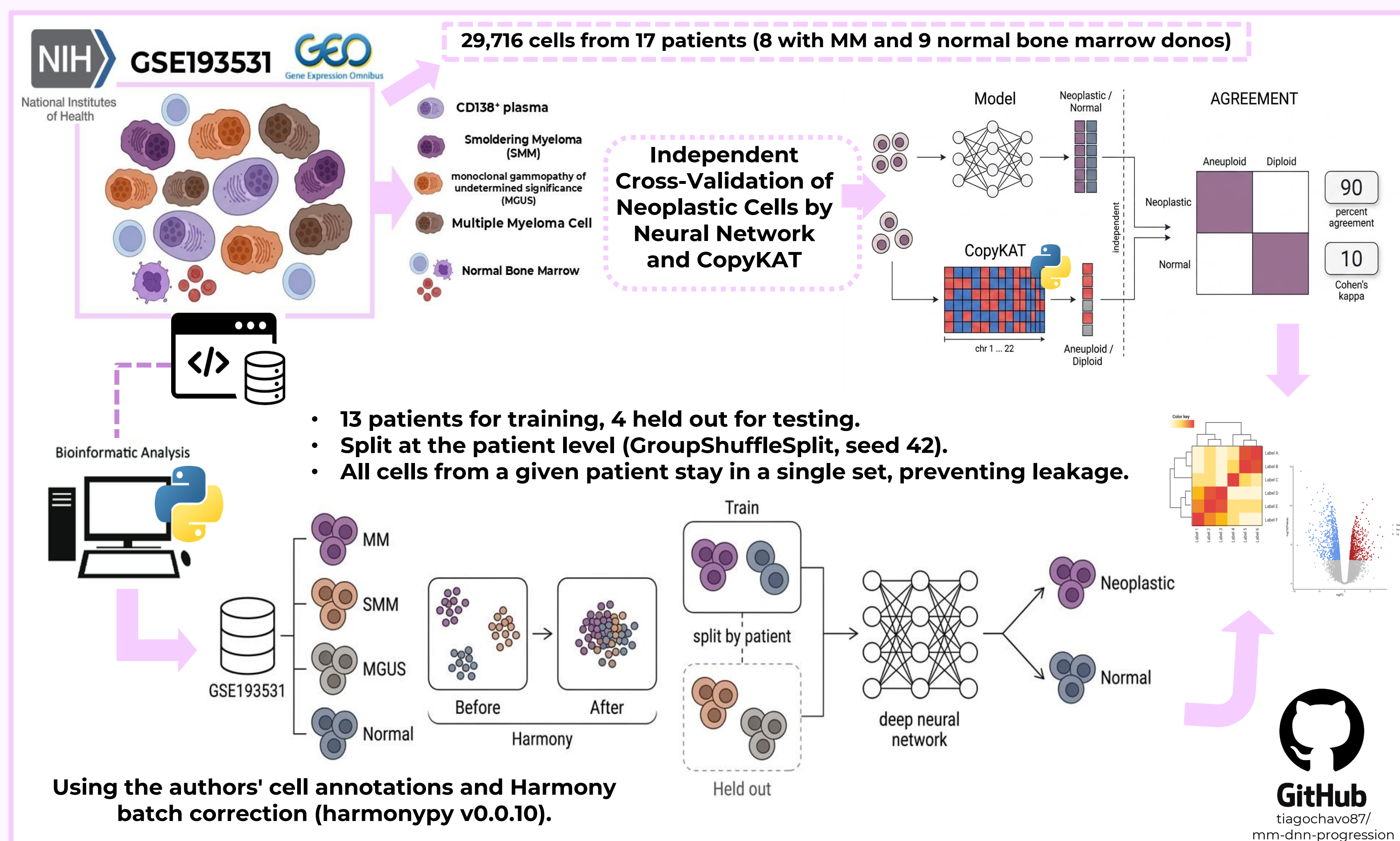


- Multiple myeloma (MM) is an incurable plasma cell malignancy.
- 188,000 new cases and 121,000 deaths worldwide in 2022.
- Cases projected to rise 70.8% by 2045.
- It is always preceded by MGUS and SMM, both asymptomatic.
- Progression to active disease is not predicted by current clinical markers.
- Who progresses, and when, remains unknown.

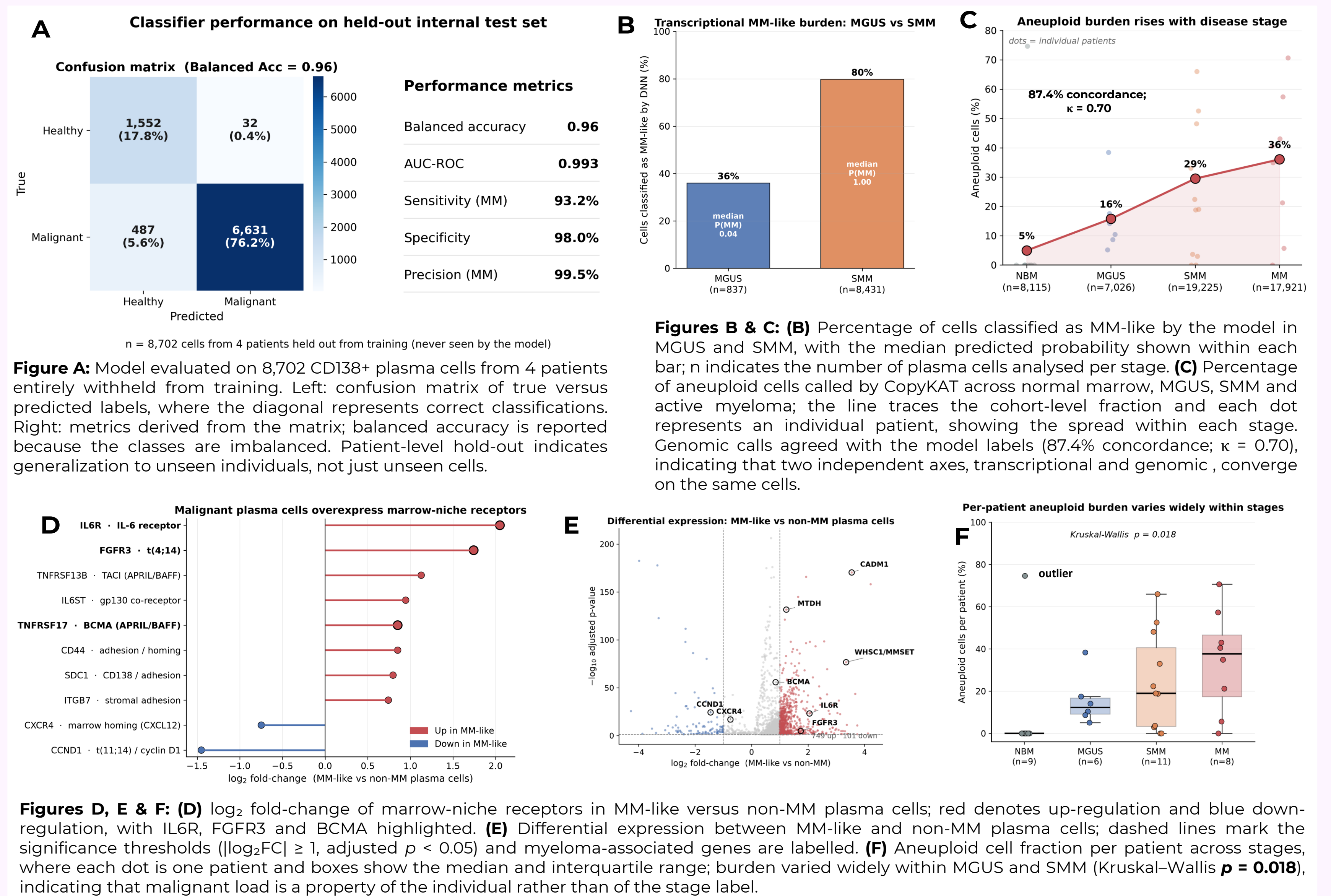
When Does Myeloma Really Begin?

We aimed to identify, at single-cell resolution, individual plasma cells within MGUS and SMM that already carry a myeloma-like transcriptional profile.

Methods



Results and Discussion



- An active-MM signature is already present in MGUS and SMM;
- Two methods agree: Deep learning + aneuploidy;
- Malignant, druggable program (WHSC1/MMSET, FGFR3, IL6R, BCMA);
- Malignant burden is a single-cell trait that varies within stages.

Conclusion

Neoplastic, aneuploid plasma cells are already present before clinical progression of MM, and their per-patient burden differs markedly among MGUS and SMM individuals, a heterogeneity that may help refine risk stratification in these stages. **Single-Cell Evidence That Transformation Precedes Diagnosis.**

References

- 1GSK. Understanding multiple myeloma: a visual guide. Behind the Science, US Edition..
2. MAURA, F.; BERGSAGEL, P. L. Molecular pathogenesis of multiple myeloma. Hematology/Oncology, v. 38, n. 2, p. 267-279, 2024
3. .MEMORIAL SLOAN KETTERING CANCER CENTER. Multiple myeloma.

Acknowledgments

